

CSE 340: Computer Architecture

[Spring 2024]

Surprise Quiz - 1 :: Set-A

Time: 5 min

Marks: 5

Name: _____

ID: _____ Section: _____

A video processing system executes a frame rendering task in 400 seconds, distributed as follows:

- Motion Estimation: 120 sec
- Transform Coding: 200 sec
- Entropy Coding: 80 sec

A new hardware accelerator improves transform coding performance, reducing its execution time by 35%. Calculate the speedup achieved using Amdahl's Law.

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Surprise Quiz - 1 :: Set-B

Time: 5 min

Marks: 5

Name: _____

ID: _____ Section: _____

A parallel computing system processes a large dataset in 500 seconds, distributed as follows:

- Data Loading: 180 sec
- Feature Engineering: 220 sec
- Model Training: 100 sec

The feature engineering phase is optimized, reducing its execution time by 30%. Calculate the speedup achieved using Amdahl's Law.